

Tushar Mishra

BTech — CSE — 2027

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Education

BTech, CSE

2023 – 2027

CGPA: 6.79

12th / Higher Secondary

CBSE / Uttar Pradesh

Science — Percentage: 70.6%

10th / Secondary

CBSE / Uttar Pradesh

Percentage : 88.4%

Skills

Languages: JavaScript, TypeScript, C++, Python

Frontend & Backend: React, Node.js, Express, Tailwind CSS, REST APIs, WebSockets

Databases & Tools: PostgreSQL, pgvector, MongoDB, Redis, BullMQ, Prisma, Git, Postman

AI/ML: RAG, FAISS, Legal-BERT, LoRA (PEFT)

Projects

TooBrain: Personal Knowledge and AI Assistant

Feb 2026 – Mar 2026

Technologies: React, TypeScript, Node.js, Express, PostgreSQL, Redis, BullMQ, RAG

- Built a JWT-secured knowledge platform that organizes five content types—PDFs, documents, web pages, YouTube transcripts, and notes—into isolated knowledge “Brains.”
- Automated a four-stage ingestion workflow—processing, chunking, embedding generation, and pgvector indexing—with Redis and BullMQ to move document processing off the request path.
- Implemented brain-scoped RAG with vector similarity search, grounding LLM responses in content selected by the user.

Legal Bail Prediction and Explainable AI System

Apr 2026

Technologies: Python, LoRA (PEFT), Legal-BERT, Flan-T5-base, FAISS, RAG

- Fine-tuned Legal-BERT with parameter-efficient LoRA to classify bail outcomes from Indian case facts, IPC sections, and crime categories.
- Built a two-stage retrieval pipeline combining FAISS dense search with cross-encoder re-ranking to surface the most relevant precedent for each case.

Internship

Backend Engineering Intern

May 2025 – Jul 2025

Center for Advanced Data and Computational Science

- Built a Node.js/WebSocket backend for bidirectional messaging across concurrent client sessions.
- Created APIs and event handlers to synchronize emotion data and generated artwork across chat rooms.
- Managed connection lifecycles and room state for reliable, low-latency message delivery.

Achievements & Responsibilities

- Published research for the COLIEE 2026 Legal Case Retrieval Task.
- Served as an Executive Committee Member of the Mrityunjay Theatre Society; acted in four major stage plays and directed two.